

4	(a)	State what is meant by <i>radioactive decay</i> .
		 [2]

		Solution: Radioactivity (or radioactive decay) is the spontaneous and random disintegration (or decay) of an unstable nucleus into a more stable one (either unstable or become more stable) with the emission of particles (and photons i.e. alpha-particle beta-particle, gamma ray photon). (particles/ radiation – can give marks if mention partial)	1 1
	(b)	A radiation detector is placed close to a radioactive source. The detector does not surround the source. Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different. Suggest two other reasons why the activity and the measured count rate may be different. 1. 2. [2]	
		Solution: • emission from radioactive daughter products • self-absorption in source • absorption in air before reaching detector • detector not sensitive to all radiations • window of detector may absorb some radiation • background radiation (not accepting a repeat mention of part of radiation is detected) Any of the above 2 points	2
	(c)	The variation with time t of the measured count rate in (b) is shown in Fig. 4.1.	

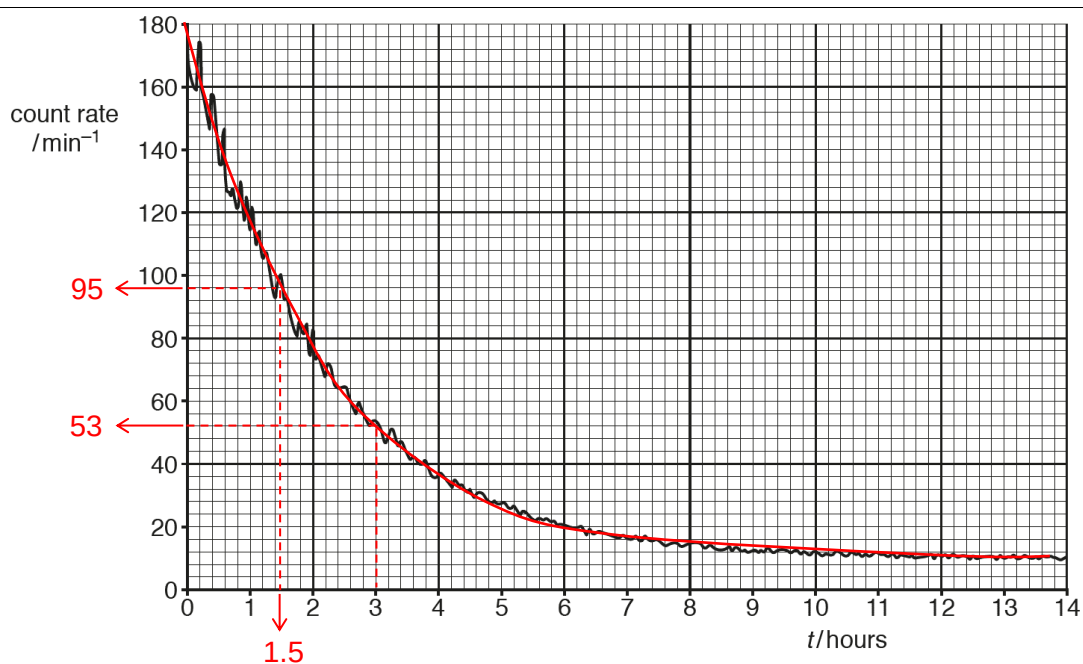


Fig. 4.1

- (i) State the feature of Fig. 4.1 that indicates the random nature of radioactive decay.

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 [1]

Solution:

Curve is not smooth or curve fluctuates or curve is jagged.

1

- (ii) Use Fig. 4.1 to determine

1. the background count rate and,

background radiation = min^{-1} [1]

Solution:

The curve is almost flat between 10 hours and 14 hours.
 background radiation = 10 min^{-1}

1

2. the half-life of the radioactive isotope in the source.

				half-life = hours [4]
			Solution: half-life determined at least twice Draw a curve of best fit to read off the values from the graph <u>First half-life</u> $C_0 = 180 - 10 = 170 \text{ min}^{-1}$ $C_1 = \frac{C_0}{2} = \frac{170}{2} = 85 \text{ min}^{-1}$ C_1 (measured) = $85 + 10 = 95 \text{ min}^{-1}$, from the graph, $t_{\frac{1}{2}} = 1.5 \text{ hour}$ (correct method of determining half life from graph) <u>Second half-life</u> $C_2 = \frac{C_1}{2} = \frac{85}{2} = 43 \text{ min}^{-1}$ C_2 (measured) = $43 + 10 = 53 \text{ min}^{-1}$, from the graph, $t_{\frac{1}{2}} = 3 - 1.5 = 1.5 \text{ hour}$ half-life = 1.5 hours – (mark for background considerations here)	1 1 2
	(d)	The readings in (c) were obtained at room temperature. A second sample of this isotope is heated to a temperature of 500 °C. The initial count rate at time $t = 0$ is the same as that in (c). The variation with time t of the measured count rate from the heated source is determined. State, with a reason, whether the heating will cause a difference, if any, in		
		1. the half-life, 		

		 2. the measured count rate for any specific time. [4] [Total: 14]	
		Solution: 1. half life : no change 1 because decay is spontaneous / independent of environment 1 2. count rate: likely to be <u>different</u> 1 because radioactive decay is random or cannot be predicted 1	

5	(a)	Explain what is meant by coherent light waves
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